Question 1

A bakery has spent all flour and is planning the flour purchases for the following 4 months, based on the orders and the historical flour cost trend. The estimated demand and price of the flour are presented for each month in the table below. In each month, the bakery can use the flour bought at the beginning of the current month and the remaining flour from the previous months. However, in order to reduce the amount of old flour used for the pastry, the bakery never buys more than the amount estimated for the current month and 30% of the estimated amount for the following month. Formulate the optimization problem that minimizes the money spent by the bakery while respecting the problem constraints. Use the decision variables $x_i, i \in \{1, \quad 2, 3, 4\}$ to model the amount of flour to buy at the beginning of each month, expressed in tonnes. Mark the box located top-left of the formulation that corresponds to this problem.

Month	1	2	3	4
Amount (t)	20	40	60	50
Price (CHF/t)	300	500	800	700

Solution 1:

$$
\begin{aligned}
&\min 300x_1 + 500x_2 + 800x_3 + 700x_4 \\
\text{s.t.}\quad & x_1 \geq 20 \\
& x_2 \geq 40 \\
& x_3 \geq 60 \\
& x_4 \geq 50 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Solution 2:

$$
\begin{aligned}
&\min 300x_1 + 500x_2 + 800x_3 + 700x_4 \\
\text{s.t.}\quad & x_1 \geq 20, \quad x_2 \geq 40 \\
& x_3 \geq 60, \quad x_4 \geq 50 \\
& x_1 \leq 32, \quad x_2 \leq 58 \\
& x_3 \leq 75, \quad x_4 \leq 50 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Solution 3: CORRECT

$$
\begin{aligned}
&\min 300x_1 + 500x_2 + 800x_3 + 700x_4 \\
\text{s.t.}\quad & x_1 \geq 20 \\
& x_1 + x_2 \geq 60 \\
& x_1 + x_2 + x_3 \geq 120 \\
& x_1 + x_2 + x_3 + x_4 \geq 170 \\
& x_1 \leq 32, \quad x_2 \leq 58 \\
& x_3 \leq 75, \quad x_4 \leq 50 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Solution 4:

$$
\begin{aligned}
&\min 300x_1 + 500x_2 + 800x_3 + 700x_4 \\
\text{s.t.}\quad & x_1 \geq 20 \\
& x_1 + x_2 \geq 60 \\
& x_1 + x_2 + x_3 \geq 120 \\
& x_1 + x_2 + x_3 + x_4 \geq 170 \\
& x_1, x_2, x_3, x_4 \geq 0
\end{aligned}
$$

Question 2

Consider the following maximization problem.

$$\begin{aligned} \max \quad & -2x_1 + 5x_2 + 3x_3 + x_4 \\ \text{s.to} \quad & 2x_1 + 4x_2 \geq 5 \\ & 8x_1 + 2x_2 + 3x_4 \leq 8 \\ & 2x_2 + 4x_3 \leq -6 \\ & |x_1| \leq 4 \\ & x_2 \geq -5 \\ & x_4 \leq 0 \end{aligned}$$

Which linear model in standard form corresponds to it?

Solution 1:

$$\begin{aligned} \min \quad & 17 + 2x_1' - 5x_2' - 3x_3' + x_4' \\ \text{s.to.} \quad & 2x_1' + 4x_2' + e_1 = 23 \\ & 8x_1' + 2x_2' - 3x_4' + e_2 = 8 \\ & 2x_2' + 4x_3' + e_3 = 4 \\ & x_1' + e_4 = 4 \\ & x_i' \geq 0 \quad i \in \{1...4\} \\ & e_i \geq 0 \quad i \in \{1...4\} \end{aligned}$$

Solution 2: CORRECT

$$\begin{aligned} \min \quad & 17 + 2x_1' - 5x_2' - 3x_3' + x_4' + 3x_5' \\ \text{s.to.} \quad & -2x_1' - 4x_2' + e_1 = -33 \\ & 8x_1' + 2x_2' - 3x_4' + e_2 = 50 \\ & 2x_2' + 4x_3' - 4x_5' + e_3 = 4 \\ & x_1' + e_4 = 8 \\ & x_i' \geq 0 \quad i \in \{1...5\} \\ & e_i \geq 0 \quad i \in \{1...4\} \end{aligned}$$

Solution 3:

$$\begin{aligned} \min \quad & -17 - 2x_1' + 5x_2' + 3x_3' - x_4' - 3x_5' \\ \text{s.to.} \quad & -2x_1' - 4x_2' + e_1 = -33 \\ & 8x_1' + 2x_2' - 3x_4' + e_2 = 40 \\ & 2x_2' + 4x_3' - 4x_5' + e_3 = 4 \\ & x_1' + e_4 = 4 \\ & x_i' \geq 0 \quad i \in \{1...5\} \\ & e_i \geq 0 \quad i \in \{1...4\} \end{aligned}$$

Solution 4:

$$\begin{aligned} \min \quad & -17 - 2x_1' + 5x_2' + 3x_3' - x_4' - 3x_5' \\ \text{s.to.} \quad & 2x_1' + 4x_2' + e_1 = 23 \\ & 8x_1' + 2x_2' - 3x_4' + e_2 = 40 \\ & 2x_2' - 4x_3' + 4x_5' + e_3 = 4 \\ & x_1' + e_4 = 8 \\ & x_i' \geq 0 \quad i \in \{1...5\} \\ & e_i \geq 0 \quad i \in \{1...4\} \end{aligned}$$

Question 3

A slack variable is:

Solution 1: a variable that is used to transform an equality constraint into an inequality constraint.

Solution 2: CORRECT a variable that is used to transform an inequality constraint into an equality constraint.

Solution 3: a variable that is used to transform a minimization problem into a maximization problem.

Solution 4: a variable that is used to transform a greater or equal inequality into a lesser or equal inequality.